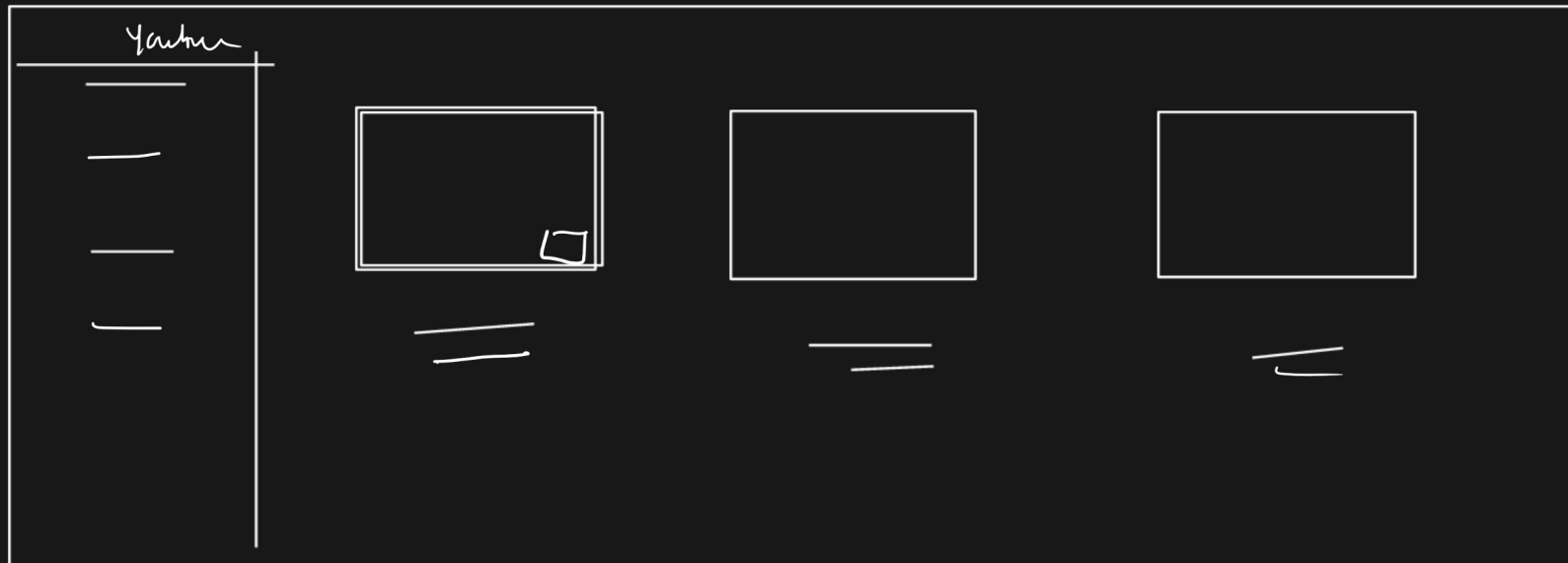


Asynchronous Programming

JS :- Synchronous Prog.
Line by line execution
Single thread program

Time
Taking
Code

```
c.log( — )  
for( let i = 0; i < 10000000; i++ )  
{  
    if( — )  
        c.log( )  
}  
c.log( — )
```

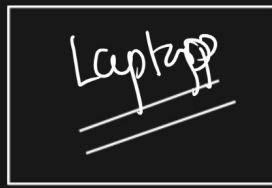


JS :- Synchronous + Asynchronous Code (Language)

JS :- Perform the asynchronous nature by using setTimeout()
setInterval(), getElementById, getElementByClassName()

* How JS run behind for synchronous as well as asynchronous in nature.

Configuration



Buy

JAVA + python + C++, C
Multithreading Prog.

- 1) Brand
- 2) Processor
- 3) Series
- 4) RAM
- 5) Internal Storage

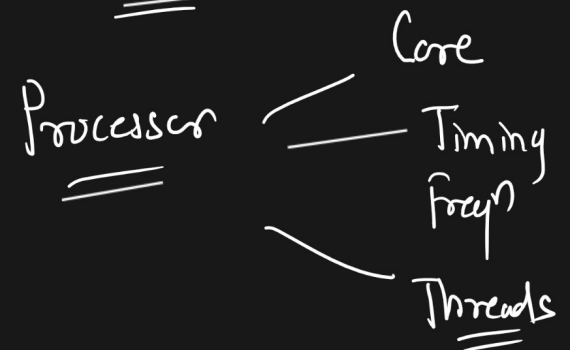
— SSD

- 6) GPU
- 7) Graphics Card
- 8) OS

9) AI Integration

10) Build concept (quality)

11) Threads



Thread :- Lightweight process



parallel
thread
small task



Multi threading
Programming

How to create a thread :- By Using Thread class $\Rightarrow$ extend
in JAVA
Thread obj = new Thread()

Runnable Interface

Python :- threading . Thread ()

Real Time Web Applⁿ - JS $\Rightarrow$ Single Threaded Program

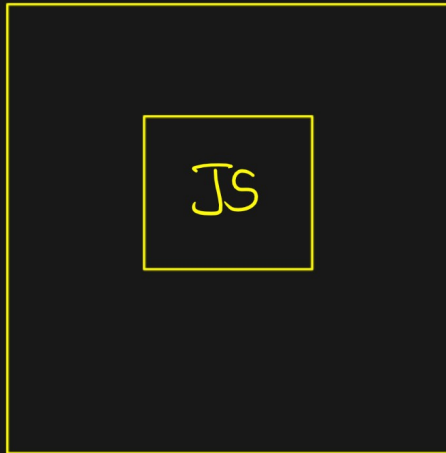
Google Drive by using JS



← upload the files (50GB)
← delete

Run Time

Environment



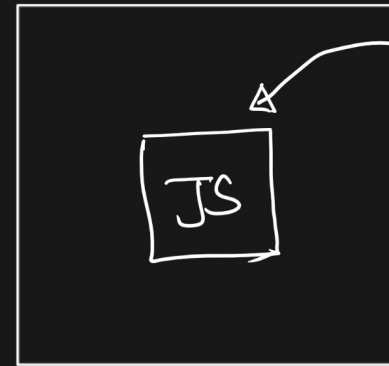
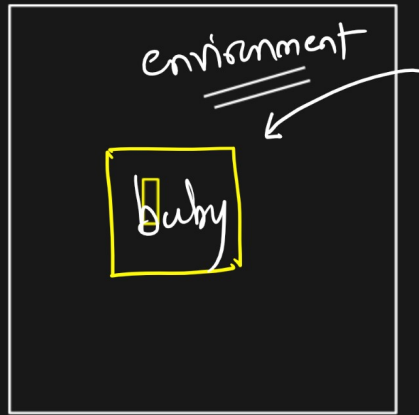
JS
JS: - (Case-1) Native Features (Functionality)
e.g. for, if, else, variables, data types

$\rightarrow$ they are execute synchronous in nature of single threaded.

(Case-2) Run Time Environment Functionality
Browser + NodeJS

$\Rightarrow$ Asynchronous in nature



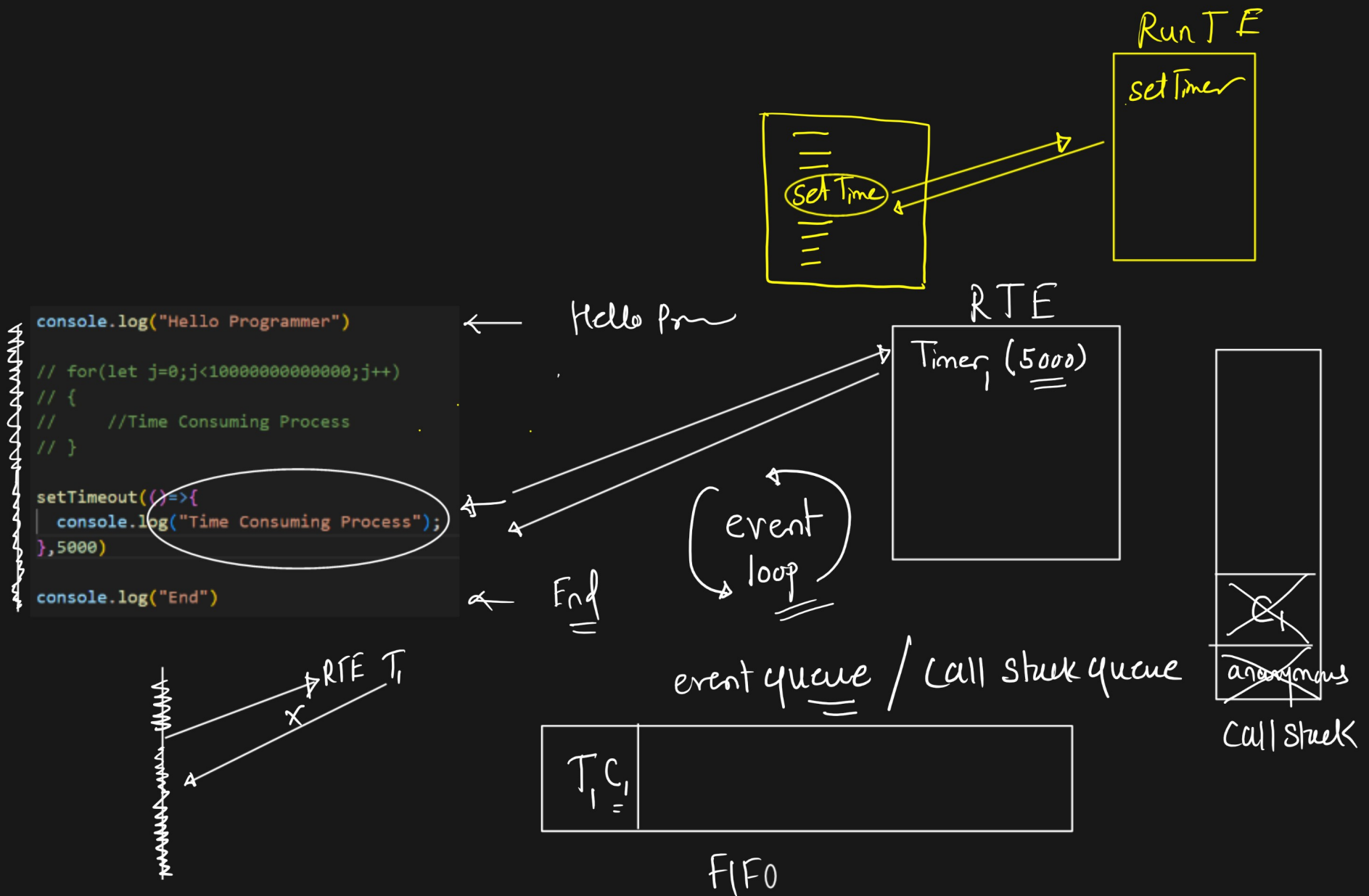


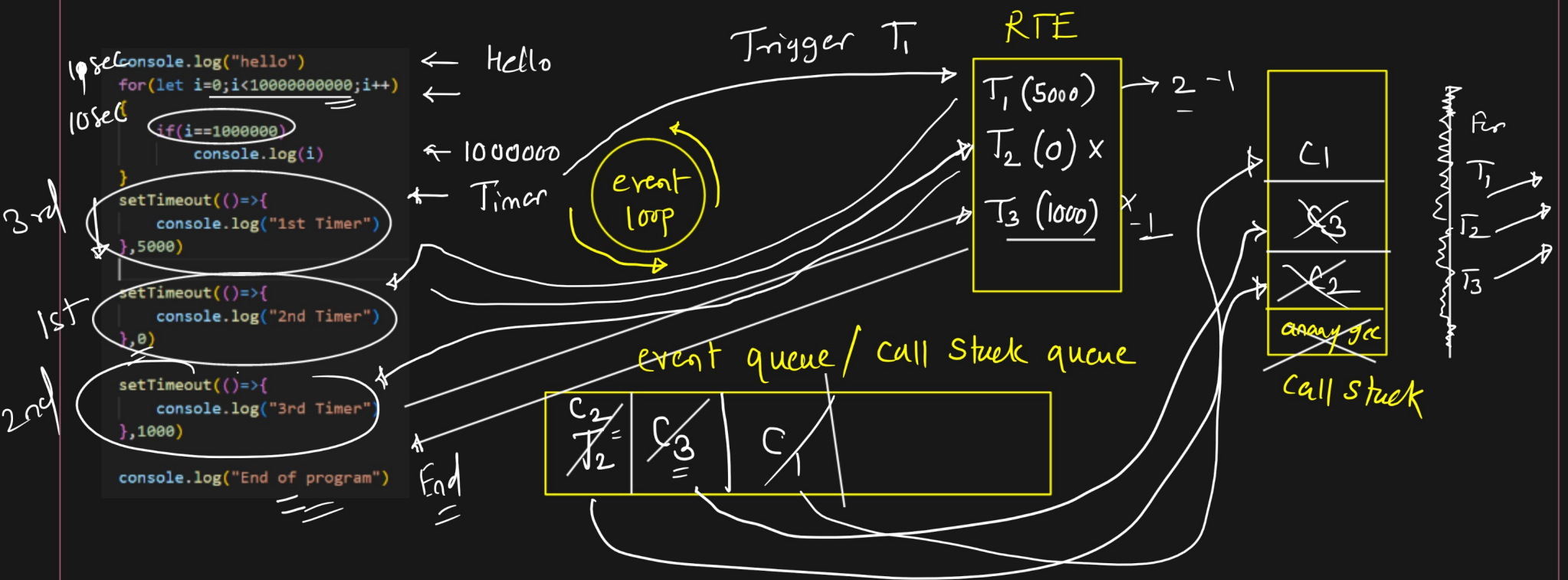
Run time
environment ↓
→ browsers
→ Node JS
→ Native JS

Run Time Environment:-

- 1) Browsers :- Timers funⁿ, DOM funⁿ
- 2) NodeJS :- File Systems

How to handle synchronous & asynchronous program behind the scene?





Interview Question:-

- 1) JS $\rightarrow$ Synchronous / Asynchronous
- 2) JS $\rightarrow$ Single Threaded / Multi Thread
- 3) How to run asynchronous program of JS?
- 4) What are the asynchronous fun'?
- 5) what is `setTimeout()` $\Rightarrow$, `setInterval`

- 6) what is callback funⁿ?
- 7) What is call stack queue / event queue?
- 8) What is event loop?